

UNIVERSITY COURSE FINDER – INTELLIGENT COURSE DISCOVERY SYSTEM

Overview

Students often struggle to identify the right online courses based on their learning goals, existing skills, preferred difficulty level, and career objectives. Traditional course search systems generally rely on keywords and may return large numbers of courses without understanding the student's actual intent.

Students may ask questions such as:

- “I want to learn data science from the basics. Which courses should I start with?”
- “I know Python and SQL. What courses should I take next to move into machine learning?”
- “I want to learn cloud computing but I don't know which prerequisite courses I need.”
- “Can you suggest a learning path from beginner to advanced data analytics?”

Instead of relying only on keyword-based search, the system should understand the student's learning goal, identify relevant courses, analyze skill gaps and prerequisites, and recommend a suitable learning path.

Build an intelligent course discovery system that allows students to enter natural-language learning goals and retrieve relevant courses along with skill-gap analysis, prerequisite information and personalized learning-path recommendations.

Tasks

Task	Task Overview	Weightage
Task 1 – Course Data Preparation & Knowledge Base	Prepare the course dataset and build the foundation for intelligent course discovery.	20%
Task 2 – Semantic Course Search & Recommendation	Enable students to search for courses using natural-language learning goals and receive relevant recommendations.	25%
Task 3 – Skill Gap, Prerequisite & Learning Path Analysis	Extend the recommendation system to identify skill gaps and create a structured learning path.	25%
Task 4 – Evaluation, Personalization & User Application	Evaluate recommendation quality and provide an application through which students can interact with the course discovery system.	20%
Task 5 – Architecture, Documentation & Demonstration	Document the complete solution and demonstrate how the intelligent course discovery system works.	10%
Total		100%

Task Details

Task 1 – Course Data Preparation & Knowledge Base

- Prepare the provided course dataset for intelligent retrieval and recommendation.
- The task should result in a searchable course knowledge base containing course descriptions, skills and relevant metadata.
- Document the preprocessing, embedding and indexing approach used.

Task 2 – Semantic Course Search & Recommendation

- Build the core course discovery capability. A student should be able to enter a natural-language learning goal and receive relevant courses.
- The system should use semantic or hybrid retrieval, support appropriate filters and provide an explanation for the recommended courses.

Task 3 – Skill Gap, Prerequisite & Learning Path Analysis

- Extend the basic recommendation system by analyzing the student's skill requirements and prerequisites.
- The solution should identify missing skills and recommend suitable courses to address those gaps.
- The system should organize the recommendations into a logical learning path from foundational to advanced courses.

Task 4 – Evaluation, Personalization & User Application

- Evaluate the quality of course recommendations, skill-gap analysis and learning paths using representative student queries.
- The solution should include a simple user interface through which students can enter learning goals and view recommendations, prerequisites, skill gaps and learning paths.
- Feedback should be captured to support improvement of recommendations.

Task 5 – Architecture, Documentation & Demonstration

- Provide the complete technical documentation and demonstrate the working solution.
- The architecture should clearly show the data-to-recommendation flow, while the design documentation should explain important technical decisions and trade-offs.
- The executable implementation and README should allow another user to set up and run the solution.

Deliverables

1. Architecture Diagram

- Include a high-level view of your system in JPEG or PDF.
- Show how data is processed from the provided dataset through to the user's course recommendation and learning-path response.

2. Design

- Articulate system design decisions taken and the trade-offs made.

3. Full Executable Code (Microservice)

- Focus on clarity and modularity.
- Provide a README explaining:
 - Project setup (how to install or run)
 - Sample usage (e.g., a student learning goal and the resulting course recommendations)

4. Panel Presentation (10 minutes)

- Demo of your working solution outlining the design and approach (8 minutes).
- QCA (2 minutes).

Dataset Name

Online Courses / University Course Dataset

Primary Link

<https://www.kaggle.com/datasets/azraimohamad/coursera-course-data>

Alternative Links

<https://huggingface.co/datasets/azrai99/coursera-course-dataset>

<https://github.com/Siddharth1698/Coursera-Course-Dataset>

<https://www.kaggle.com/datasets/siddharthm1698/coursera-course-dataset>

Format

CSV

Key Fields

Course name, course description, skills, difficulty level, rating, organization and other relevant course metadata.

Reason

The course dataset provides information required to support semantic course discovery, skill matching, prerequisite analysis and personalized learning-path recommendations.